

Part 2. Data Cleaning Guidance

1 Introduction

This document provides complete guidance for the cleaning of `trip_path` and `waypoint` datasets. All procedures align with the official FHWA technical reference TO 22-274 Task 3.1 User Manual v6.0, available at: https://drive.google.com/file/d/17i0PV_IyQRM8M4xrbUg4HARdgWKkj62K/view?usp=sharing.

The input datasets referenced in this guidance correspond to the official project dataset package available at: https://drive.google.com/file/d/1_s_s4RLNbEwbKE6QUEGA_8eyfrIEcWiz/view?usp=sharing.

The purpose of this document is to provide a reproducible and scalable workflow to harmonize timestamps, eliminate duplicated or noisy observations, remove erroneous data, and prepare the cleaned data for downstream data fusion in Part 3.

2 Objectives

The cleaning tasks focus on:

- Time zone alignment and timestamp standardization using ISO format.
- Duplicate removal for both trip path and waypoint datasets.
- Noise filtering in spatial and temporal dimensions.
- Outlier and error data removal using physical and statistical rules.
- Optional waypoint based map matching preparation for link level integration.

3 Datasets

- `trajs.csv`: vendor B trip path dataset.
- `waypoint.csv`: vendor A connected vehicle waypoint dataset.
- `SegmentId_to_link.csv`: mapping between SegmentId and GMNS link identifiers.

4 Data Cleaning Procedures

This section consolidates all cleaning related procedures into a single workflow, including timestamp harmonization, duplicate removal, noise filtering, outlier detection, and error record removal.

4.1 Timestamp Standardization

Trip path timestamps are typically recorded in UTC based ISO strings, while waypoint timestamps are stored as Unix epoch values. Consistent with the FHWA manual guidelines, all timestamps should be converted to the correct local time zone for the study corridor and standardized into a uniform ISO format. Records with missing or invalid timestamp information should be removed to ensure temporal integrity before further processing.

4.2 Duplicate Removal

Duplicate records may appear due to platform specific sampling patterns, transmission delays, or device level buffering. All duplicated observations should be removed to ensure temporal consistency and to avoid inflating probe density in downstream analyses. The specific duplicate identification strategy is left to the implementer, provided that the final dataset maintains one valid observation per unique event. The percentage of removed duplicates should be reported for data quality assessment.

4.3 Outlier Removal

Outlier detection can follow several different strategies depending on the characteristics of the data and the study corridor. Multiple methods are acceptable, and implementers may choose the approach best suited to their dataset. Eligible options include:

- **Rule based filtering:** remove missing speeds, non positive speeds for unrealistic durations, or speeds that exceed physically reasonable thresholds such as corridor dependent maximum speeds.
- **Statistical outlier detection:** identify anomalous speed values using approaches such as interquartile range filtering, z score thresholds, or distribution based tail analysis.
- **Temporal consistency checks:** detect abrupt or unrealistic transitions in speed or position relative to the expected sampling interval or vehicle movement pattern.

Trip path speed in `CrossingSpeedKph` should be converted to mph before outlier evaluation. Selected methods should be applied consistently across the dataset.

4.4 Error Data Removal

- Remove trip path records whose `SegmentId` cannot be mapped to GMNS links.

- Remove waypoint points that fall outside the routable network or cannot be associated with any road segment.
- Remove rows with missing required fields such as speed, coordinates, or timestamps.

4.5 Optional Waypoint Based Map Matching

Waypoint map matching is an optional step that enhances link level analysis in Part 3. Instead of implementing a custom algorithm from scratch, students are encouraged to use existing Python based map matching tools such as:

- gotrackit: <https://pypi.org/project/gotrackit/>
- mapmatcher4gmns: <https://pypi.org/project/mapmatcher4gmns/>

These tools can take cleaned waypoint trajectories and a GMNS formatted network as inputs, perform geometry and topology based map matching, and return link level trajectories.

The matched outputs should include, at a minimum, link identifiers, timestamps, and derived speeds so that they can be directly integrated into the link level fusion procedures in Part 3.

5 Deliverables

The final deliverables include:

- Local time aligned trip path and waypoint datasets formatted using ISO timestamps.
- Spatially and temporally filtered observations that meet the required data quality standards.
- Cleaned datasets with duplicates, outliers, and invalid records removed.
- Optional map matched waypoint datasets suitable for link level integration.
- A summary of data quality metrics documented before and after the cleaning process.
- Fully reproducible cleaning scripts accompanied by clear and comprehensive documentation.

These deliverables form the standardized inputs for the subsequent data fusion procedures in Part 3.